Vishnu Vardhan

Full Stack Engineer

9182177915 repanavishnuvardhan67664@gmail.com LinkedIn GitHub Portfolio Bengaluru, Karnataka, India

Professional Summary

Full Stack Engineer with nearly **1 year of internship experience** building production-grade web applications end-to-end — responsive React/Next.js frontends, scalable REST APIs (Flask, FastAPI, Node.js/Express), and optimized relational and NoSQL databases. Track record of measurable performance gains: **30–40% lower latency**, **3× API throughput**, **35% faster page loads**, and zero critical CVEs across live client deployments. Strong foundation in system design, application security, and cloud deployment, with working experience integrating AI/LLM features into full-stack products.

Experience

The Developers Arena — Hyderabad, Telangana, India
Full Stack Developer Intern — Nov 2025 – May 2026

- Architected **6+ production-grade REST APIs** with **Flask and FastAPI** using **modular MVC design patterns**, enabling seamless third-party integrations and reducing average response latency by **30%** across live client deployments.
- Engineered **optimized MySQL schemas** with normalized table design, composite indexing, and stored procedures — cutting query execution time by **40%** and sustaining reliable performance under high-volume concurrent read/write workloads.
- Scaled **API throughput by 3×** by integrating **Apache Kafka event streaming** for real-time data ingestion and implementing **Redis caching, Celery async task queues, and connection pooling** — sustaining **sub-200ms p95 response times** under **500+ concurrent request** load tests.
- Secured backend infrastructure against **OWASP Top 10** attack vectors by integrating **JWT/OAuth2 auth flows, RBAC, AES-256 encryption, and parameterized queries** — achieving **zero critical CVEs** across all API-facing services and blockchain endpoints.
- Built **end-to-end backend pipelines** integrating third-party **AI/LLM services (OpenAI, Gemini)** with existing REST APIs — delivering **document Q&A, semantic search, and automated reasoning** features to production without disrupting core services.
- Refactored monolithic backend into **Dockerized microservices** with **Kafka-driven inter-service messaging and RESTful contracts** — reducing deployment downtime by **50%** and enabling **zero-downtime rolling updates** across high-traffic components.
- Automated **data preprocessing pipelines** using **Pandas and NumPy** across **50K+ record datasets** — standardizing workflows and reducing prep time by **35%**.
- Sustained **95%+ sprint velocity** across **8+ Agile/Scrum cycles** using **Git flow, GitHub PRs, and Jira** — coordinating across backend, frontend, QA, and DevOps squads. **View Certificate**

Infotact Solutions — Bengaluru, Karnataka, India
Full Stack Developer Intern — Sep 2025 – Dec 2025

- Designed and delivered **2+ production-ready full-stack MERN applications** (**React.js, Node.js, Express.js, MongoDB**) featuring **responsive UI/UX** and **RESTful backend architecture** supporting real-world client workflows.
- Engineered **secure, scalable REST APIs** with **JWT-based authentication, Role-Based Access Control (RBAC), and server-side input validation** — ensuring **zero unauthorized access** across all frontend-backend communication channels.
- Optimized **application performance by 35%** through **MongoDB indexing strategies, query optimization, aggregation pipeline tuning, and React component refactoring** — reducing page load time and improving user experience.
- Implemented **CI/CD-ready development workflows** using **Git Flow branching, GitHub Pull Requests, and structured peer code reviews** — enforcing **DRY principles, modular architecture, and SOLID design patterns** across the codebase.
- Collaborated within an **Agile/Scrum team** across **planning, development, testing, and deployment phases** — consistently delivering features on schedule with **high code quality and minimal defect rate**. **View Certificate**

Skills

Programming Languages: Python, JavaScript, TypeScript
Frontend Development: React.js, Next.js, HTML, CSS, Responsive UI Design
Backend Development: Flask, FastAPI, Node.js, Express.js, REST APIs, Authentication, Scalable Backend Services
Databases: MySQL, SQL, NoSQL (MongoDB), Vector Databases
System Design & Security: MVC Architecture, Microservices, OWASP Top 10, JWT/OAuth2, RBAC, AES-256 Encryption
Cloud & DevOps: AWS, Docker, Kubernetes, CI/CD Pipelines, Deployment Automation, Kafka, Redis, Celery
Tools & Platforms: Git, GitHub, Postman, VS Code, Jira, n8n, Jupyter Notebook, Vercel, Netlify
AI/ML Exposure: LangChain, LangGraph, LLM Integration, RAG, Prompt Engineering, Scikit-learn
Methodologies: Agile, Scrum, Software Development Life Cycle (SDLC)

Projects

N45 Agentic Builder | *Internship Project — Next.js, React, TypeScript, FastAPI, Python, MySQL, AWS*

- **Business problem:** enterprises lacked a unified visual platform to build, orchestrate, and monitor multi-agent AI workflows — relying instead on fragmented scripts and manual integrations.
- **Architecture:** contributed to a production-ready **Enterprise AI Agent Development Platform** featuring a drag-and-drop **Workflow Builder (React Flow)** and **multi-agent orchestration** for parent–sub-agent collaboration.
- Worked on **workflow validation, versioning, autosave, deployment, and real-time execution monitoring** for reliable end-to-end workflow lifecycle management.
- Supported implementation of **JWT authentication, RBAC, multi-tenant workspace architecture, and encrypted credential management** for enterprise-grade security.
- Assisted in connecting enterprise tools (**Google Workspace, Slack, GitHub, Jira, Notion, REST/GraphQL APIs, webhooks**) with **analytics dashboards and audit logging** for full observability.
- **Result:** platform enabled enterprises to build and scale AI agent workflows without custom infrastructure — cutting integration overhead across providers and tools.

AI-Based Vehicle Speed Detection System | *HTML, CSS, JavaScript, Python, Flask, OpenCV, Deep Learning, TensorFlow/PyTorch, MySQL*

- **Problem:** Traditional traffic monitoring systems rely on **expensive hardware (e.g., radar guns, sensors)** and manual supervision, making them costly and less scalable.
- Built a **full-stack AI system** to estimate vehicle speed from video, eliminating specialized hardware.
- Implemented **real-time detection, tracking, and optical flow** for accurate speed estimation.
- Achieved **˜95% accuracy** with a low average error of **±7 km/h** across diverse traffic conditions.
- Developed **real-time speed visualization dashboard** for monitoring vehicle speeds dynamically.
- Built a **Flask-based backend** with secure **user authentication (login/register)** and REST APIs.
- Integrated **MySQL database** to store user data and vehicle speed records.
- Designed a **responsive frontend** with video upload and interactive UI using HTML, CSS, and JavaScript.
- Engineered the system for **scalability, anomaly detection, and vehicle re-identification**. **GitHub Repo**

Online Fee Management System | *HTML, CSS, JavaScript, Python, Flask, MySQL, AI (LangChain, LLMs, RAG, Vector DB)*

- **Business problem:** manual fee collection at **Aditya Degree College** caused delays in fee tracking, reporting, and student query handling.
- **Architecture:** designed and shipped a full-stack AI-powered platform (**Flask REST APIs, MySQL, responsive UI**) to digitize and automate fee operations end-to-end for a live institutional client.
- Delivered **secure admin authentication**, **automated fee tracking**, and **digital receipt generation**.
- Integrated **LLMs + RAG** for intelligent fee queries, summaries, and administrative assistance — reducing turnaround on routine student queries.
- **Result:** cut manual workload and improved fee-management efficiency with a real-time analytics dashboard.
- **Next steps scoped:** multi-agent automation layer (**n8n**) to orchestrate fee reminders, receipt workflows, and admin task handling. **GitHub Repo**

Education

Aditya Degree College — B. Kotha Kota
Bachelor of Computer Applications (BCA); CGPA: 9.05/10 — 2026

Awards & Achievements

- Awarded **Best College Website Developer** by Aditya Degree College for excellence in website development (2025).